

Urinary & Reproductive Systems

The kidney does not make urine. It filters everything, then decides what to keep.

BY THE END OF THIS PAGE YOU SHOULD BE ABLE TO

- Name the kidney's jobs beyond making urine
- Describe filtering, reabsorbing, and secreting, and say where each happens
- Explain how ADH and aldosterone change urine and blood pressure
- Explain why kidney failure upsets so many unrelated systems
- Follow the hormones that control the reproductive cycle

1 A filter that keeps almost everything

Your kidneys filter about 180 liters a day, and you make one or two liters of urine.

That is the design in one sentence. Instead of picking out what to remove, the kidney removes nearly everything small enough to pass, then **takes back what it wants**.

It sounds wasteful until you see the advantage. Anything unfamiliar, like a drug or a toxin the body has never met, gets filtered out by default and is left behind unless there is a transporter to save it.

Beyond removing waste, the kidney also controls **fluid volume and blood pressure, salt balance, and acid balance**. It also releases hormones that drive red cell production and activate vitamin D.

THINK OF IT THIS WAY

Not a strainer that catches trash. A warehouse that dumps everything onto a belt and puts back only the items on its list. Anything not on the list leaves.

2 Three steps, in order

- 1 **Filtering.** Blood pressure forces water and small particles out of the capillary. Cells and large proteins stay behind, so finding either in urine means the filter is damaged.
- 2 **Reabsorbing.** Sugar, amino acids, most sodium, and most water are taken back. Normally *all* the sugar is recovered, which is why sugar in urine means the blood level was high enough to overwhelm the transporters.
- 3 **Secreting.** Extra hydrogen ions, potassium, and many drugs are added into the tube. This is a second chance to remove things that were not filtered well.

Urine is what is left after all three. Filtering is blind. The intelligence is in reabsorbing and secreting.

3 The two hormones that change the answer

Switch them on and off and watch the urine change.

ADH, ALDOSTERONE AND THE URINE THEY PRODUCE

Urine volume	1.5 L/day
Urine concentration	600 mOsm/kg
Sodium	balanced
Potassium	balanced
Blood volume & pressure	steady

Normal hormone levels give a moderate volume of moderately concentrated urine, with sodium and potassium in balance. Press either button to see what changes when the kidney is told to conserve or to let go.

- **ADH** lets the last part of the tube take back **water**, so urine gets small and concentrated. It is released when you are dehydrated.
- **Aldosterone** makes the tube take back **sodium**, and water follows the sodium. It also pushes potassium out into the urine. It is released when blood pressure or volume

falls.

Both raise blood volume, but by different routes. ADH saves water alone. Aldosterone saves salt and takes water with it. That is why one dilutes the blood and the other does not.

WHY THE NURSE CARES

Water pills work by blocking sodium reabsorption. Sodium stays in the tube, water stays with it, and volume and blood pressure fall. Because sodium handling is tied to potassium, checking potassium is part of using them safely.

4 Why kidney failure touches everything

Six jobs in one organ means six problems when it fails.

LOST JOB	WHAT THE PATIENT GETS
Removing waste	Waste builds up: nausea, confusion, exhaustion
Fluid control	Fluid overload, swelling, breathlessness, high blood pressure
Potassium handling	Rising potassium, which threatens the heart rhythm
Acid balance	The blood becomes acidic
Red cell hormone	Anemia, even with normal marrow and normal iron
Activating vitamin D	Low calcium and weakening bone, the loop from Module 7

None of those look like a urine problem at the bedside. They make sense only once you see the kidney as a hormone and balance organ that happens to produce urine.

5 Reproductive control — the same chain again

Hypothalamus, pituitary, target gland. You already know the shape of this.

The hypothalamus releases a starter hormone. The pituitary answers with **FSH** and **LH**. The gonads respond by making sex cells and sex hormones, which feed back on the first two steps.

- **In the male**, FSH supports sperm production and LH drives testosterone. Testosterone keeps sperm production, muscle, and bone going, and it feeds back to quiet the chain. That is why taking testosterone from outside shuts down the body's own production and can reduce fertility.
- **In the female**, the same two hormones drive a monthly cycle. FSH ripens a follicle, which makes estrogen. Estrogen rebuilds the uterine lining and, once it is high enough for long enough, triggers a surge of LH. That surge causes **ovulation**. What is left of the follicle then makes progesterone, which keeps the lining ready.

With no pregnancy, progesterone falls, the lining is shed, and the cycle starts again. With a pregnancy, the embryo makes a hormone that keeps progesterone up. That hormone is exactly what a pregnancy test looks for.

Notice the exception. The LH surge is a rare case of **positive feedback**. Rising estrogen calls for more LH, not less, until ovulation ends it. Like clotting and labor, it is used where the body wants an event finished, not a value held steady.

6 Check yourself

Five questions. The wrong answers explain themselves, so they are worth reading.

1. Why does the kidney filter about 180 liters a day only to take almost all of it back?

- a) It is wasteful, and the body tolerates it
- b) Filtering everything means anything unfamiliar leaves unless it is deliberately saved
- c) It is the only way to control blood pressure
- d) Reabsorption needs a large starting volume

2. Sugar shows up in a patient's urine. What does that suggest?

- a) The filter is damaged
- b) Blood sugar was high enough to overwhelm the transporters
- c) The kidney stopped filtering sugar
- d) Aldosterone is low

3. A dehydrated patient passes a small amount of dark urine. Which hormone is responsible?

- a) Aldosterone
- b) ADH
- c) Parathyroid hormone
- d) Insulin

4. A patient in kidney failure is anemic even though iron and bone marrow are normal. Why?

- a) The kidney filters out red cells
- b) The kidney normally makes the hormone that tells the marrow to build red cells
- c) Urine destroys red cells
- d) Anemia has nothing to do with the kidney

5. What directly triggers ovulation?

- a) A drop in progesterone
- b) A surge of LH, caused by estrogen staying high
- c) A rise in FSH alone
- d) The lining being shed

Answer key

1. **Filtering everything means anything unfamiliar leaves unless it is deliberately saved**

Correct. Drugs and new toxins have no transporter to save them, so they leave.

2. **Blood sugar was high enough to overwhelm the transporters**

Correct. All filtered sugar is normally taken back, so its appearance means the transporters were swamped.

3. **ADH**

Correct. ADH lets the tube take water back, so urine gets small and concentrated.

4. **The kidney normally makes the hormone that tells the marrow to build red cells**

Correct. Marrow and iron are fine, but the signal to produce is missing.

5. **A surge of LH, caused by estrogen staying high**

Correct, and it is a rare case of positive feedback: rising estrogen calls for more LH, not less.

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Joyce University of Nursing and Health Sciences

Course resources by **Dr. Vidya Garg**